Q3 NO CALCULATOR ALLOWED Q3

Answer QUESTION 3 PARTS A and B on this page.

PART A

$R'(1) = \frac{R(2) - R(0)}{2 - 0} = \frac{100 - 90}{2}$ words/min^2

PART B

$R(t)$ is differentiable and therefore continuous

$R(0) = 90$ $R(10) = 162$ $90 < 155 < 162$. $\therefore$ by the IVT, there must exist at least one c such that $R(c) = 155$ on the interval $0 \le c \le 10$.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Answer QUESTION 3 PARTS C and D on this page.

PART C

$$\int_0^{10} R(t)\,dt \approx \frac{1}{2}\left[2(90+100) + 6(100+150) + 2(150+162)\right]$$

PART D

$$W(t) = -\frac{3}{10}t^2 + 8t + 100$$

$$\int_0^{10} W(t)\,dt = -\frac{1}{10}t^3 + 4t^2 + 100t \Big|_0^{10}$$

$$= \left[\frac{-1}{10}(10)^3 + 4(10)^2 + 100(10)\right] - [0] \text{ words}$$

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